Desk Companion

Plan your day by stacking it on your real desk, and see overcommitment before you live it.

Productivity track | New Experience | Target: Best New Productivity Experience

AT A GLANCE

Track / division	Productivity (habit tied to morning routine and wind-down) / New Experience
Primary prize	Best New Productivity Experience ($70,000; runner-up $35,000)
Fallback special award	Best Agentic Interaction ($25,000) or Best Reason to Come Back ($25,000)
Session length	Morning ritual about 5 minutes; wind-down about 3 minutes
Room requirement	A desk or table; tap-to-place fallback on any flat surface
Core tech	Plane detection, anchors, hand tracking and grabbing, IndexedDB, a small serverless AI function
Biggest risk	'Why put on a headset to plan my day?'
Signature moment	The stack rising past the red capacity line

The one-line pitch

A mixed-reality daily planner where tasks have physical size. Each task becomes a tile whose thickness is its time estimate, and you sort the tiles by hand into zones on your real desk. Stack more than your day can hold and the pile rises past a glowing capacity line: overcommitment you can see and feel.

SECTION 01

The idea

To-do apps are lists, and lists hide the thing that actually hurts: you planned nine hours of work into a five-hour day. In mixed reality, tasks can have physical size. Put too much on your desk and you see the pile overflow.

Desk Companion is not a to-do list in VR. It is your workload made physical, sitting on your actual desk. It is designed around two short, recurring moments: a 5-minute morning plan and a 3-minute evening wind-down.

> **The core mechanic**
>
> Every task is a tile with real thickness: a 30-minute task is a thin tile, a 3-hour task a thick block (for example 1 cm per 30 minutes, so an 8-hour day is a 16 cm stack). A translucent capacity line at the Now zone is set from the hours you actually have today. Stack past it and the top tiles glow red and wobble. You feel the overcommitment instead of reading about it.

SECTION 02

How a day works

Morning ritual (about 5 minutes)

1. **The tray appears.** Put on the headset at your desk. The app finds the table surface and places a soft glowing day tray on it, anchored to your real desk.
2. **Brain dump.** Dictate or type what is on your mind. The agent splits it into task tiles, each with a time estimate. Tiles drop onto the desk.
3. **Sort by hand.** Pinch-grab tiles into three zones drawn on the desk: Now, Later and Drop. Tiles snap into place and stack automatically.
4. **Capacity line.** As the Now stack grows it approaches the line set by today's available hours. Past it, the top tiles glow red and wobble.
5. **Agent nudge.** A short suggestion appears beside the stack, such as 'Now is 1.5 h over. These two are lowest priority. Move them to Later?' One pinch accepts.

During the day (optional)

Pull a tile toward your chest and it becomes a focus card, with a timer floating above the desk while the other tiles dim.

Wind-down (about 3 minutes)

- Sweep finished tiles into a done jar on the corner of your desk. The jar fills across days and weeks and never resets.
- For unfinished tiles, flick left to push them to tomorrow, or drop them in Drop.
- For each finished tile, a quick slider asks roughly how long it took (optional, one pinch).
- The agent gives a one-line recap: 'You planned 6 h and did 4.5 h. Your estimates on writing tasks run about 40% low.'

SECTION 03

Improvements over the first draft of this concept

Original plan	Change	Why
Free-form stacking by hand	Tiles snap into zones and stack automatically; the pinch only chooses the zone.	Precise stacking with hand tracking is fiddly and a failed drop ruins the demo.
Agent notices dependencies like 'waiting on a colleague's review'	Nudges only use information the user actually gave: estimates, priority, capacity, and explicitly stated dependencies.	Judges will test the agent; it should never claim knowledge it cannot have.
Estimation-bias learning as a headline feature	Show it in the video as a feature that grows over weeks; do not headline it.	It needs weeks of real data that will not exist by the deadline.
Voice input assumed	Voice if available; otherwise a companion phone page (scan a code) or quick-pick templates.	Speech recognition inside a WebXR session in Quest Browser is unverified.
Plane detection assumed	Tap-to-place on any flat surface is a first-class path.	A table is only found if the user labelled it during Space Setup.

SECTION 04

Why passthrough and the real room matter

- **The content lives on your desk.** The tray, zones and stacks sit on your real work surface, at real scale. With passthrough off the concept breaks, which is what the judges want.
- **Physical size carries meaning.** Height on your desk is a real measurement you can compare with the objects around you: 'today's plan is taller than my coffee mug' is an intuitive signal.
- **The done jar is a permanent object in your room.** It sits on your desk corner every day, filling up.

SECTION 05

The first five minutes

Time	What happens
0:00	Passthrough. A soft light traces the edge of the desk and a day tray settles onto it.
0:20	A prompt: 'What is on your mind today?' The user speaks or uses the fallback input.
1:00	Tiles drop onto the tray one by one, each labelled with a title and time estimate.
1:30	Three zones glow: Now, Later, Drop. The first tile pulses; the user pinches it and drops it in Now. It snaps and stacks.
3:00	The Now stack approaches the capacity line; the line brightens. One more tile and it glows red and wobbles.

Time	What happens
3:30	The agent nudge appears; one pinch moves two tiles to Later and the stack settles under the line with a satisfying sound.
4:30	Done. 'See you this evening.' The first run is the tutorial, done with the user's own tasks.

SECTION 06

The agent

The AI makes the experience faster, but the experience stands without it, which satisfies the take-it-away test: tiles can be created manually, sorted physically, checked against capacity, and swept into the jar with no AI at all.

Agent job	Input	Output	Visible as
Split the brain dump	Free text	A list of tasks, each with a title, estimate in minutes, category and optional stated dependency	Tiles dropping onto the tray
Capacity nudge	Tasks in Now, their estimates and priorities, today's hours	One suggestion with a short reason	A floating card beside the stack; one pinch to accept
Split a big task	One large task	Two or three smaller tasks	The tile cracks into smaller tiles
Evening recap	Planned versus done, reported durations	One sentence	Text above the done jar
Estimation bias (over weeks)	History of estimated versus reported times per category	A per-category correction factor	Estimates gently adjusted, with a note

Implementation

- A small serverless function (for example on Vercel) holds the API key so it never reaches the headset.
- The function calls a fast Claude model (for example Claude Haiku 4.5) with structured output (a fixed JSON schema), so the app always receives valid task data.
- The app sends only the text needed for the request and stores all tasks locally in IndexedDB. No accounts and no server-side storage.
- If the network or AI is unavailable, the app falls back to manual tile creation with preset durations (15 min, 30 min, 1 h, 2 h).

SECTION 07

Progression and reasons to come back

System	Description
The done jar	Fills with finished tiles across days and weeks and never resets. The main visible growth.
Streaks	Consecutive days with both a morning plan and a wind-down.
Honest estimates	Over weeks, the app learns your estimation bias per category and shows how your planning improves.
Weekly shelf	At the end of each week, the jar's contents compress into a labelled layer, so months of work are visible as strata.

SECTION 08

Accessibility

- Voice input where available, with typing and preset alternatives.
- Large tiles and generous snap zones for limited fine motor control.
- Capacity warnings use motion, height and pattern, not only red colour.
- Captions for all audio; high-contrast mode for tile text.

SECTION 09

Art direction and sound

Soft, tactile and calm: frosted-glass tiles with clear typography, gentle glows for zones, a warm glass jar. The overflow moment is the one dramatic beat: red glow, wobble and a low hum. Sounds are soft clicks for snapping, a glassy chime for tiles entering the jar.

SECTION 10

Technical architecture (IWSDK)

Feature	Used for
Plane detection	Finding the desk (the scaffold already enables it).
Hit test	Tap any surface to place the tray when no table is found.
Anchors	Keeping the tray and jar in place across days (persistence in Quest Browser needs checking).
Grabbing and hand tracking	Picking up and dropping tiles; flick and sweep gestures.
UI (UIKit)	Tile text, nudge cards, recap text.
IndexedDB	All task data, history and jar contents.

Feature	Used for
Serverless function	The agent calls.

System	Responsibility
DeskSetupSystem	Places the tray and jar; saves the anchors.
TaskTileSystem	Creates tiles from task data; sets thickness from the estimate.
ZoneSnapSystem	Snaps dropped tiles into zones and restacks them.
CapacitySystem	Compares stack height with the capacity line; triggers the overflow effects.
AgentSystem	Calls the serverless function and turns responses into tiles and nudges.
FocusSystem	The focus card and timer.
JarSystem	Done jar, streaks and weekly strata.

SECTION 11

Risks and mitigations

Risk	Severity	Mitigation
'Why a headset to plan my day?'	High	Target people who already work in a headset; make the overflow moment striking in the first 20 seconds of the video.
Productivity is a crowded idea	High	Lead with the capacity line and the done jar, not feature breadth.
Voice input unavailable in WebXR	Medium	Phone companion page or presets; test in week 1.
Fiddly grabbing	Medium	Snap zones and automatic stacking.
Anchor persistence unreliable	Medium	Re-detect the desk each session; store jar contents in IndexedDB regardless.
Agent overreaches	Medium	Structured output; nudges based only on known data.

Week-1 checks

1. Does the Web Speech API work inside an immersive WebXR session in Quest Browser?
2. Do persistent anchors survive closing and reopening the app?
3. Does plane detection return a labelled table in your own room?
4. Can hand tracking reliably pinch a 6 cm tile lying on a desk?

SECTION 12

Seven-week build plan

Week	Dates	Goal	Exit criterion
Week 1	29 Sep - 4 Oct	Platform checks (speech, anchors, planes, pinch); tray placement	All four week-1 questions answered
Week 2	5 - 11 Oct	Tiles with thickness; zones; snapping and stacking	Manual tiles can be created and sorted
Week 3	12 - 18 Oct	Capacity line and overflow effects; agent function for brain dump	The overflow moment works on device
Week 4	19 - 25 Oct	Agent nudge; first-run flow; IndexedDB persistence	A new user completes the morning ritual unaided
Week 5	26 Oct - 1 Nov	Wind-down: done jar, flick to tomorrow, recap; focus card	Content complete
Week 6	2 - 8 Nov	Performance; playtests over several days; accessibility options	Testers use it on multiple days
Week 7	9 - 15 Nov	Polish; record video; write submission	Video and text ready
Ship	16 - 18 Nov	Final checks; submit on 17 Nov	Submitted early

SECTION 13

Scope

Must ship	Nice to have	Cut
Desk detection and tray; tiles; three zones; capacity line; done jar; agent brain dump	Focus mode; estimation-bias learning; streak visuals; weekly strata; sound design	Calendar integrations; accounts; sync across devices

SECTION 14

Demo video plan (under 3 minutes)

Time	Beat	Shot
0:00 - 0:15	Hook	Tiles piling past the red line on a real desk, wobbling.
0:15 - 1:30	Morning ritual	Real-time brain dump, tiles dropping, sorting by hand.
1:30 - 2:00	The nudge	The agent's suggestion; one pinch; the stack settles below the line.
2:00 - 2:40	Wind-down	Sweeping tiles into the jar; flicking unfinished ones to tomorrow.
2:40 - 2:55	Coming back	Montage of the jar filling across days.

SECTION 15

Submission draft

Name	Desk Companion
Tagline (under 140 characters)	Plan your day by stacking it on your real desk. See overcommitment before you live it.
Track / division	Productivity / New Experience
Target launch	Web release Q1 2027

Description (first draft)

To-do lists hide the problem that hurts most: planning more than a day can hold. Desk Companion gives every task physical size on your real desk.

Each morning, tell Desk Companion what is on your mind. An AI agent turns it into task tiles whose thickness is their time estimate. Sort them by hand into Now, Later and Drop. When the Now stack rises past your capacity line, the top tiles glow red and wobble, and the agent suggests what to move. Each evening, sweep finished tiles into a done jar on your desk that fills up over weeks.

We built it with the Immersive Web SDK using plane detection, anchors and hand grabbing, with a small serverless function for the agent. Everything works without the AI too: you can create, sort and complete tasks entirely by hand.

SECTION 16

How it maps to the judging criteria

Criterion (25% each)	Evidence in Desk Companion
Innovation and Creativity	Workload as physical height; overcommitment made visible and tangible.
Experience Design	Two recurring rituals; first run is the tutorial; content sits low and central.
Technical Implementation	Plane detection with fallback, anchors, grabbing, an AI agent with structured output.
Polish and Presentation	The overflow shot is a strong hook; calm, tactile visuals.

Hard requirement and design guidelines

Test	What the rules ask	How this idea passes
Hands-first (hard requirement)	Complete the entire experience without ever pairing a controller.	Every action is a pinch, grab, sweep or flick.
Airplane seat test	Every interaction works within a two-foot radius, seated.	Everything happens on the desk within arm's reach.

Test	What the rules ask	How this idea passes
One bus stop test	Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less.	Morning is 5 minutes, wind-down 3 minutes; each is complete on its own.
Take it away test	Remove any third-party service and a project still stands.	Remove the AI and you can still create tiles, sort them, see capacity and fill the jar.

Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.
- **Wrapper rule:** the AI is an accelerator, not the product. Make sure the video shows manual tile creation too.

SECTION 17

Honest weaknesses

- The biggest risk is the question 'why put on a headset to plan my day?'
- Productivity apps are a common idea; others will build task managers too.
- Voice input in Quest Browser needs checking.
- No multiplayer, so the Social award is out of reach (which is fine).

SECTION 18

Scorecard and verdict

Dimension	Score / 10	Comment
Originality	7	The capacity line is original; the category is not.
Only possible in MR	8	Physical size on a real desk is the point.
Works in untested rooms	8	One surface plus a fallback.
Solo feasibility in 7 weeks	7	Moderate; the agent adds an external dependency.
Prize ceiling	$70k	Productivity track; Agentic Interaction as a fallback.

Verdict

A solid, credible Productivity entry built around one strong mechanic. Choose it if you prefer the Productivity track or want to compete for the Agentic Interaction award; focus everything on the overflow moment and the done jar.